

EnLearning

| Making Education Accesible |

Meet the Team



Shivani
(team lead)



Samantha



Peggy



Kathy
(mentor)



Tiffany



Aagneyo



Manassés

Agenda

Topics Covered

Identifying language barrier challenges

Defining challenges in education and ideas on app

Implementation of features in app

Research Investigation / Proposition

Defining the Problem

In a world of **globalization**, it is about impossible for anyone to completely escape effects such as the development of lingua francas. Since English has increasingly popularized into the **universal language**, the need for interpretation has never been more prominent. When diving into specific barriers, there has been a notable language gap between **teachers and exchange students**. In 2021-2022, there were nearly **1 million exchange students** in the United States, most of whom were not fluent in English.

Education should be **accessible** to everyone, but unfortunately, many students can not **comprehend** concepts that are not in their native language. It is hard to teach them complex courses, such as physics, by itself, let alone in a completely different language. While teachers have tried their best to help exchange students adapt, many of them **lack the resources**. Even so, many current resources are not fully able to teach the students since they don't have **modern smart technology**, for example, AI.

Background

Our project was originally designed to create a solution to teach **non-native speakers** language, primarily English, so they can **communicate** more easily and efficiently.

However, **research** lead us to notice the **difficulties** that **international/ exchange/ transfer students** face when they move from a **non-English speaking country to an English speaking country**. From difficulties understanding **cultural** norms, as expressed in research done by Tang I. Heng in 2017, to a more obvious **language barrier**.

One of our members brought up an issue they witnessed in their **classroom**: a foreign exchange student struggling to keep up with class due to the language gap. After doing further research and **brainstorming**, we decided to shift our love for **language** into creating a solution that might help **bridge the gap in understanding a non-native English student in an academic setting**.

Proposition

Implementing a tool that could **break the language gap** between students and teachers worldwide could create **better education conditions** for every app user (it doesn't matter the language or location of the client).

The app would **translate** the conversations between teachers and students around the globe and **documents** and **audio** and **video** classes. This connection could help students have more **accessible** contact with **educational materials** and soften the lack of teachers that some communities have been facing daily.

Is important to say that the app will **not just translate** like other tools (such as google translator), the app will use **AI** to build other **unique** features for our users apart from pure translation.

Hypothesis

If teachers and students have access to a resource where they can ensure a comprehensive understanding of the concepts being taught, we can bridge the language barrier gap. This resource will take the form of an app designed to serve the needs of both teachers and students. Specifically made to accommodate those with concerns and desires to enhance their knowledge, this application will include study materials, **digital** tools, and other great resources. These tools can help facilitate a deeper level of **understanding** and **communication**, enabling both **educators** and learners to overcome **linguistic** challenges and **enhance** learning experiences.

Data collection methods

Researching AI

As **AI** was largely significant to our solution, we did some research.

We researched the different **types** and purposes of AI to see which one would fit best for our app.

We found certain types of AI, like Chat-GPT-already used for educational purposes- would be sufficient when compared to other types.

Survey

We created a **survey** with a variety of questions that assessed opinions over our app.

With over **110** responses, the majority being students, we gained **beneficial** data.

This data would later on help us decide which features to add in order for us to stand out among other apps with a similar purpose.

Prototype

We used **Glide** to create a **prototype** of our app.

Through this prototype, we had a **visual** image of what it would look like.

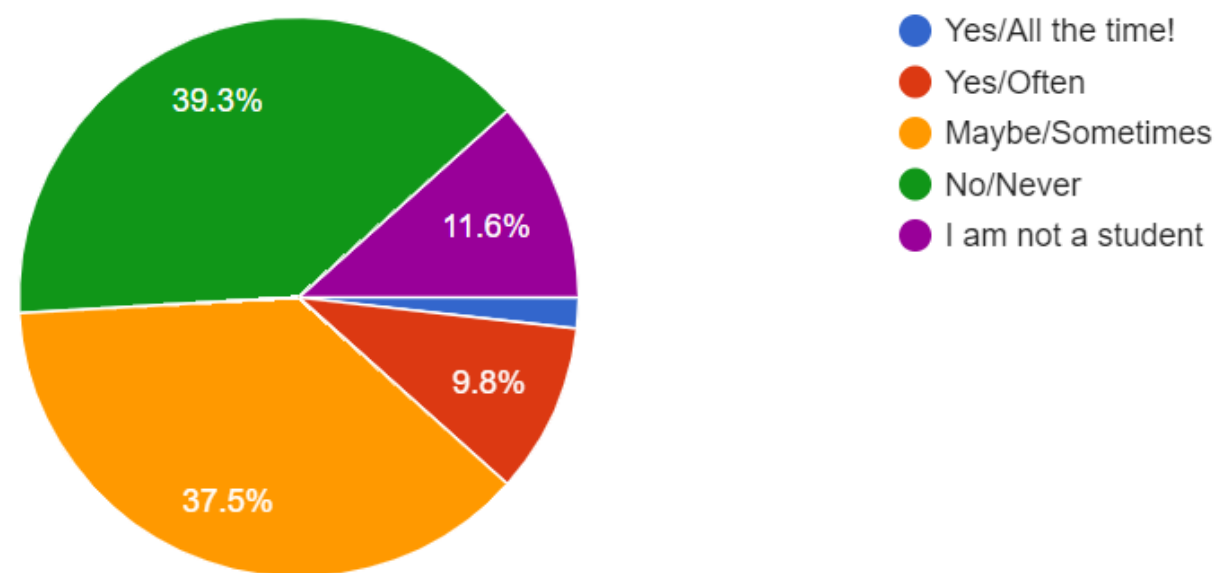
Along with this, it helped us imagine how future students could **navigate** the app and it's features. This was a way to implement all our **data** into a visual.

Results

We received more than **110** responses and found that **70%** of our responders were students and 60 % are not ESL/ELL learners.

If you're a student, do you ever find difficulty understanding what's being taught in class due to a language barrier?

112 responses



If you're an educator, do you sometimes struggle teaching something to a student whose first language isn't English?

112 responses



We found that 37.5% of people found that they **sometimes** experience **difficulty** in learning concepts and the **majority** of educators sometimes **struggle** to teach students that isn't fluent in English

Results

As for feedbacks, here are what some of the responders said:

Not able to understand how well this would be realistically as we have seen through examples of google translate that the exact translation is difficult to obtain and it is much harder to understand once each word is translated compared to one just explaining in their mother tongue

Translating using softwares like google translate isn't always accurate and so the text written by students or teachers might translate into something else, leading to miscommunications. If you haven't already considered and solved this problem, you might want to think about how to avoid it.

ease make the live captions accurate and also adding a feature that helps non-native English speakers learn English would be beneficial as they could then start understanding the educators themselves

While providing comprehensible input to English Language Learners is extremely important, especially in the early stages of learning the language, I think it's also important to facilitate learning the new language. If you rely too heavily on translation features, it could slow down language acquisition. Could you add in scaffolds to help the student learn English, or maybe an ability to adjust the level of assistance offered to the student based on their current level of fluency?

The majority of the people found all the tools (resources, live caption, communication...) provided in the app to be useful.

I think it is a wonderful idea, but most of these features are already readily available via other means, such as Google Translate. Since most teachers are already using Google tools, it is likely easier to get them to use them instead of a new app. I have had to teach other teachers how to do things like uploading documents/slides into Google Translate because they are not aware of this. The solution is most likely bringing awareness of the tools that are available to educators, rather than creating a new app.

Analyzing Results

Based on our testing, a significant of respondents, **over 92%**, believe that our app will be quite **useful**. However, there were concerns raised regarding **potential competition** from apps like Google Translate, as well as, translation disparities.

57.8% of respondents said that their first language was not English. With approximately **42% are facing difficulties as a result of language barriers**. Notably, almost all educators found challenges in teaching students whose first language is not English.

80% of respondents found our **key features** to be useful, showing that we can **follow through** with making this app.

Majority of the people rated the communication feature (8-10) higher than the other features (resource feature: 7-10)

Solution

Our final solution aims to help exchange students and teachers. The purpose of our app is to **close the language barrier in order to allow the students to learn efficiently**. Specifically, our team focused on **English** for reasons stated previously. Additionally, we wanted to make sure our app **stood out** from other translating apps. Hence, we included several **unique** features:

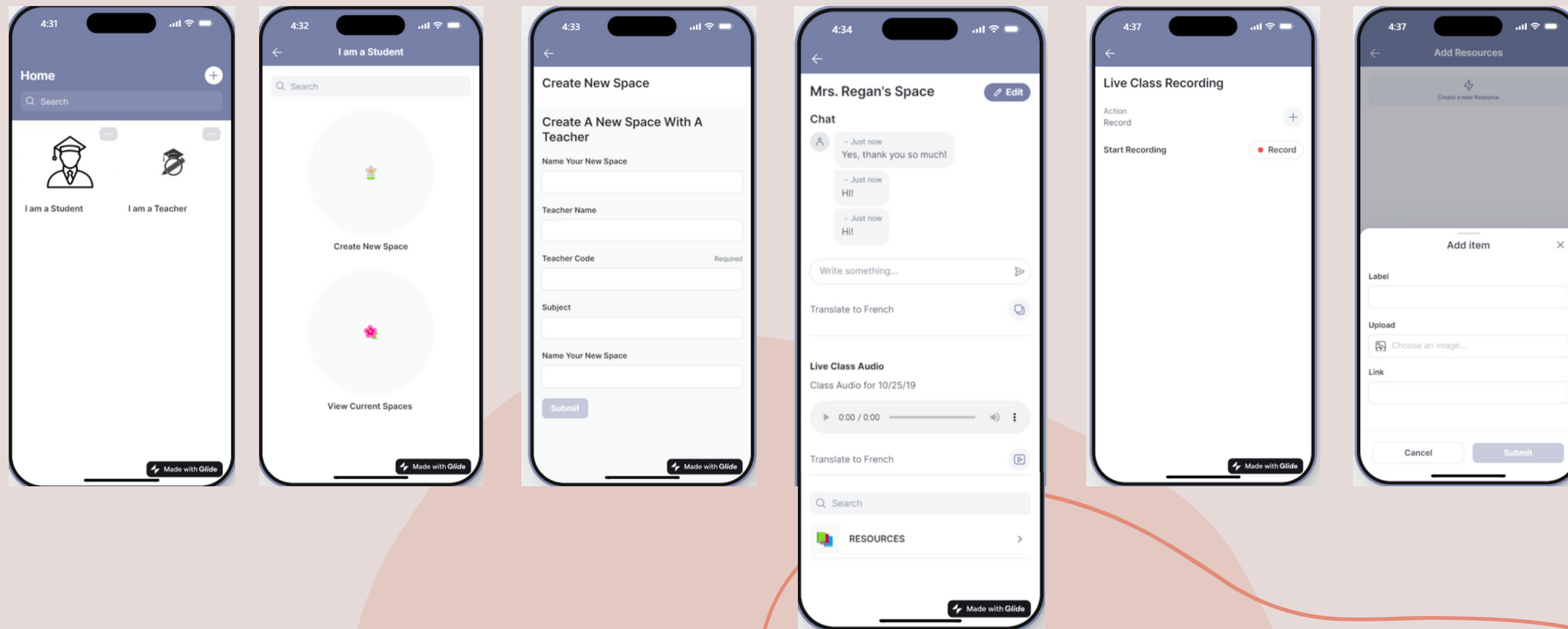
1. **Live captions**- a feature that can be used live in class when the teacher turns on recording and the student can access the live captions of what is being said (in both English and their native language)
2. **Direct communication platform** between teachers and students.
3. **Teaching material**, such as flashcards that can be uploaded by teachers and viewed by students in English and their native language.

Our goal is less of learning the language, but more of being able to **interpret educational material through translations**.

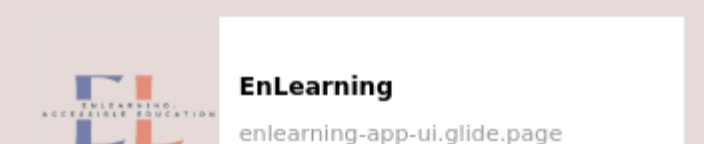
It's significant to denote that as a part of the Cognitive Classrooms challenge, we heavily relied on AI for our features.

App Wireframes

We prototyped our solution using the tool Glide.



(scan the QR code/ click the link to visit our app prototype!)



[click here!](#)

Implementing the solutions

The app has two different tabs, one for the students and the other for the teachers:

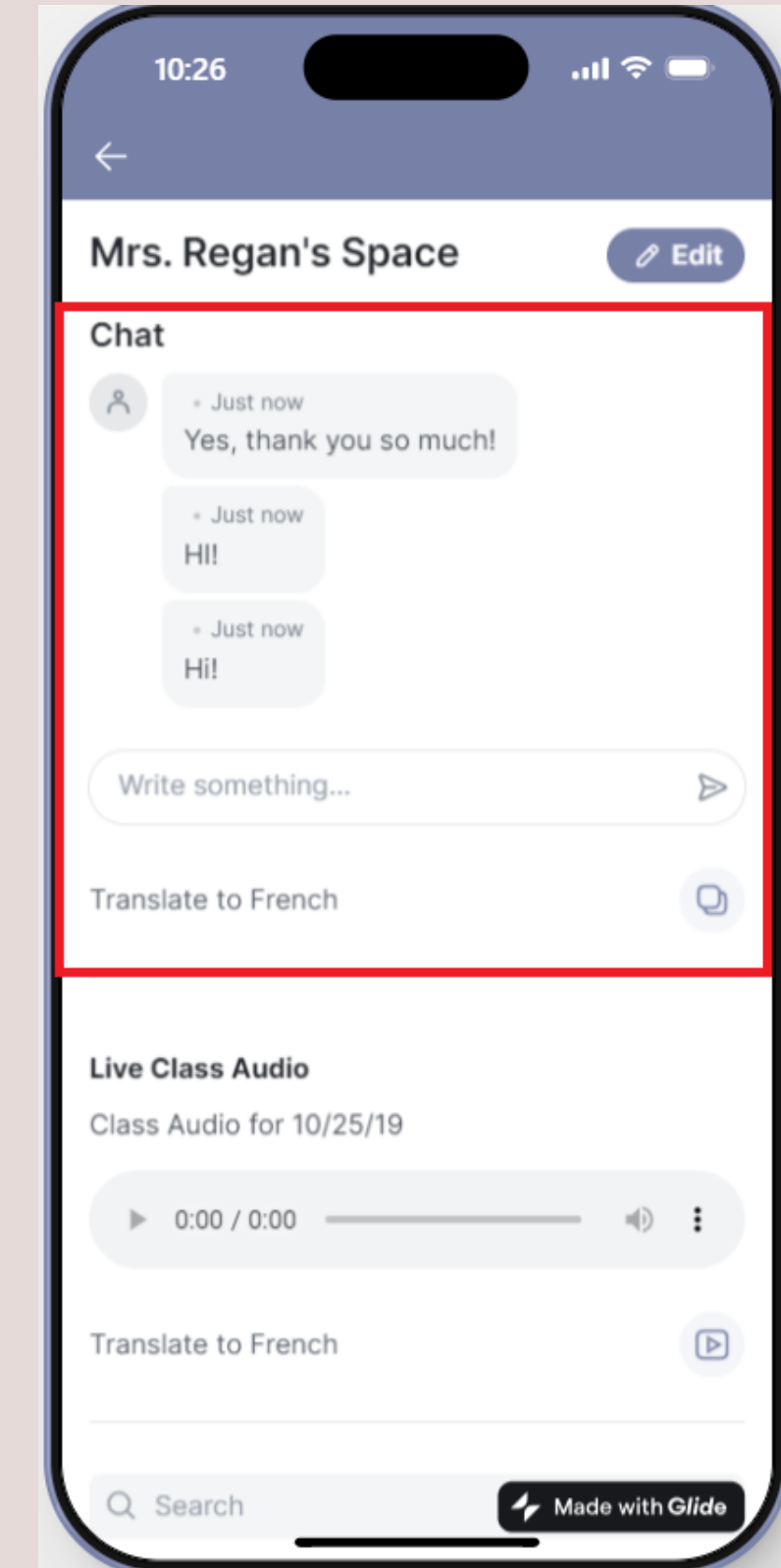
On the student's part, we can **visualize** the possibility of adding a new teacher to your **network** and the current spaces of the **connections** that you made with your teachers. In these "current spaces", there is each contact, and when you click, you access the tab where you can chat, listen to audio, and see the resources the teacher sent you.

On the teacher's part, we can visualize the same options of adding new connections and visualizing the current spaces. When you click in some student space, you can see the tools to **chat**, send **resources**, and **record** audio.

Communication Feature

We plan to have a simple **direct messaging** feature in our app that will allow a teacher and student to be able to communicate outside of the google classroom. This allows the student to ask any additional doubts and get them clarified by the teacher. Any explanations the teacher sends will have the option of being **translated** into the **native language** of the student if they wish to view it that way.

We believe that adding this feature will make it easier for the student to **understand concepts** better and also give teachers a way to send **extra materials** to the student.



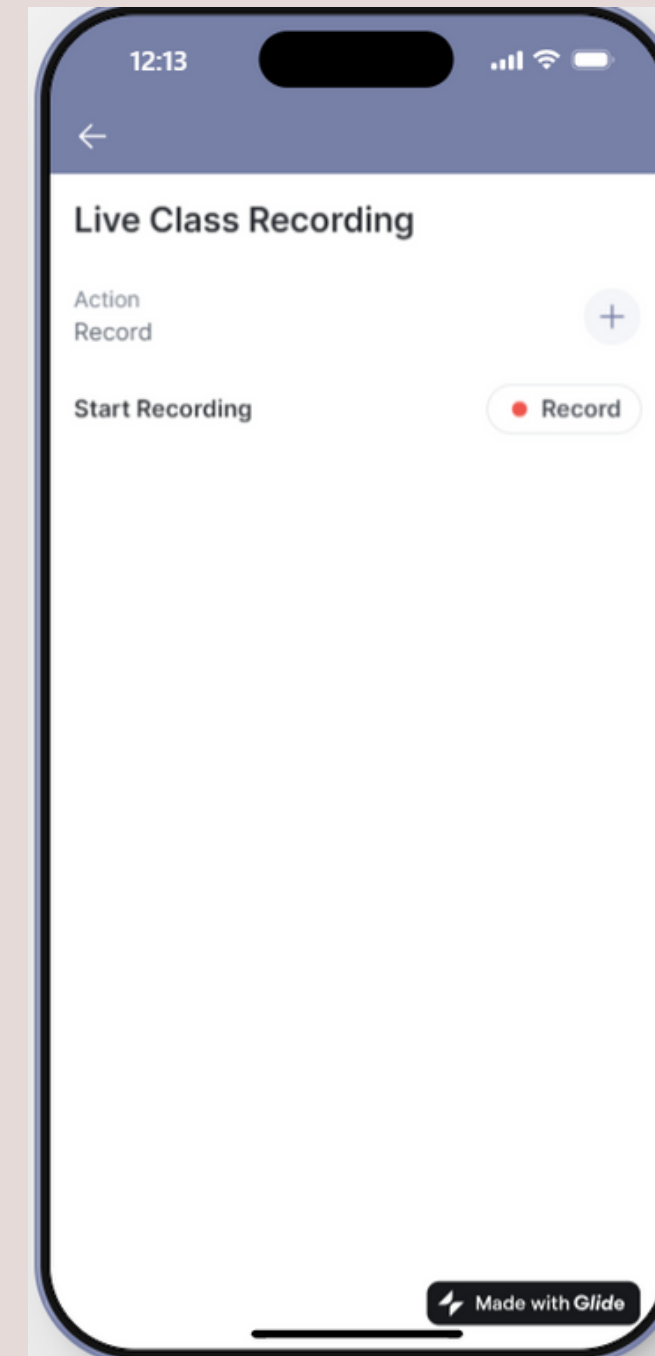
**The chat feature
for the student.**

Live Class Option

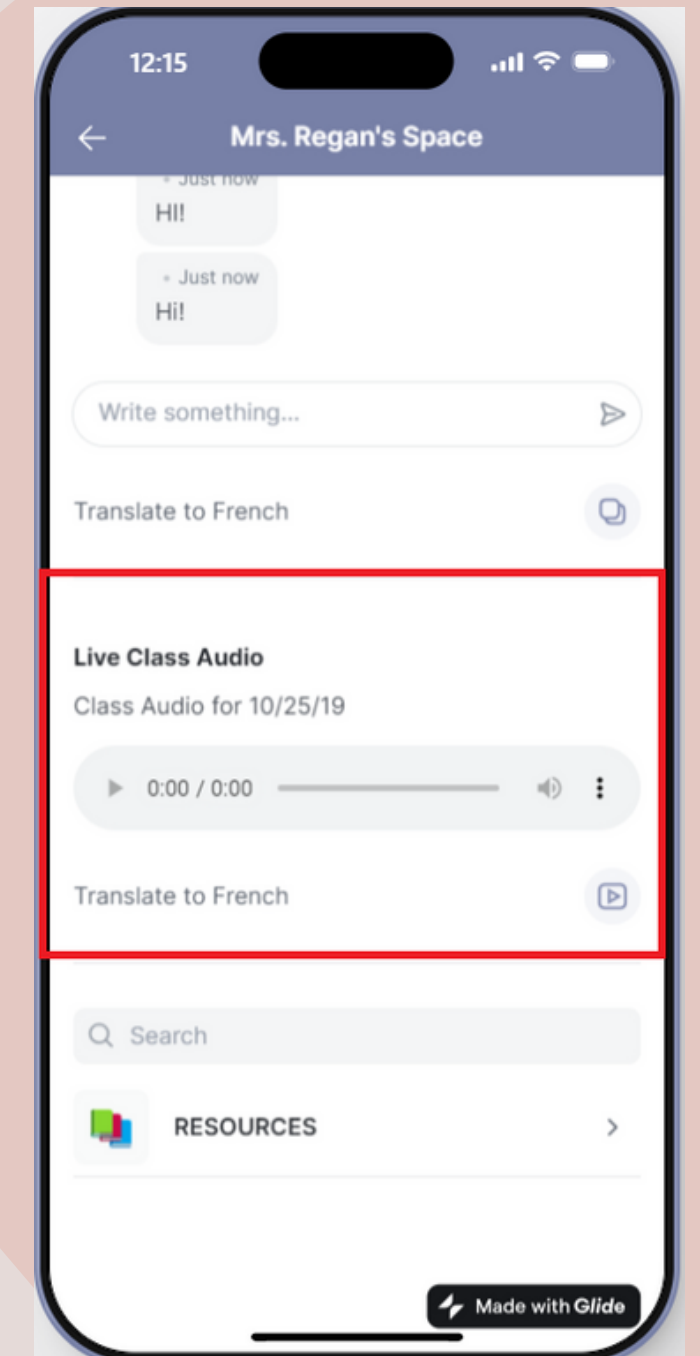
In order to ensure that students can fully understand what is being taught in class immediately, we think that it would be helpful for students to get **text captions** of class lectures.

During online classrooms, this was an option available through tools like Google Meet, but we want to make this available for **offline** classrooms as well.

The teacher can generate a code and start recording, and the student can access the code to join the recording and get captions. This all can happen **in class during lectures**.



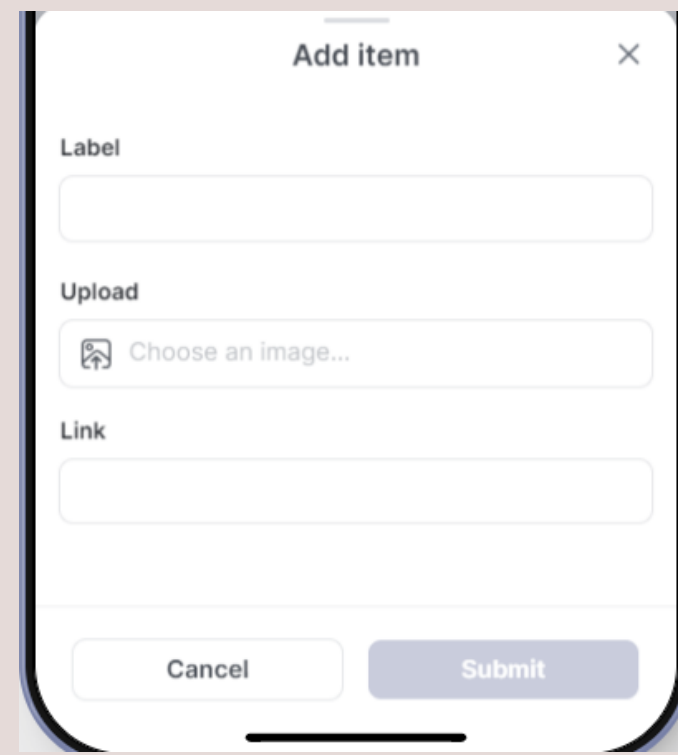
The **teacher's** recording side.



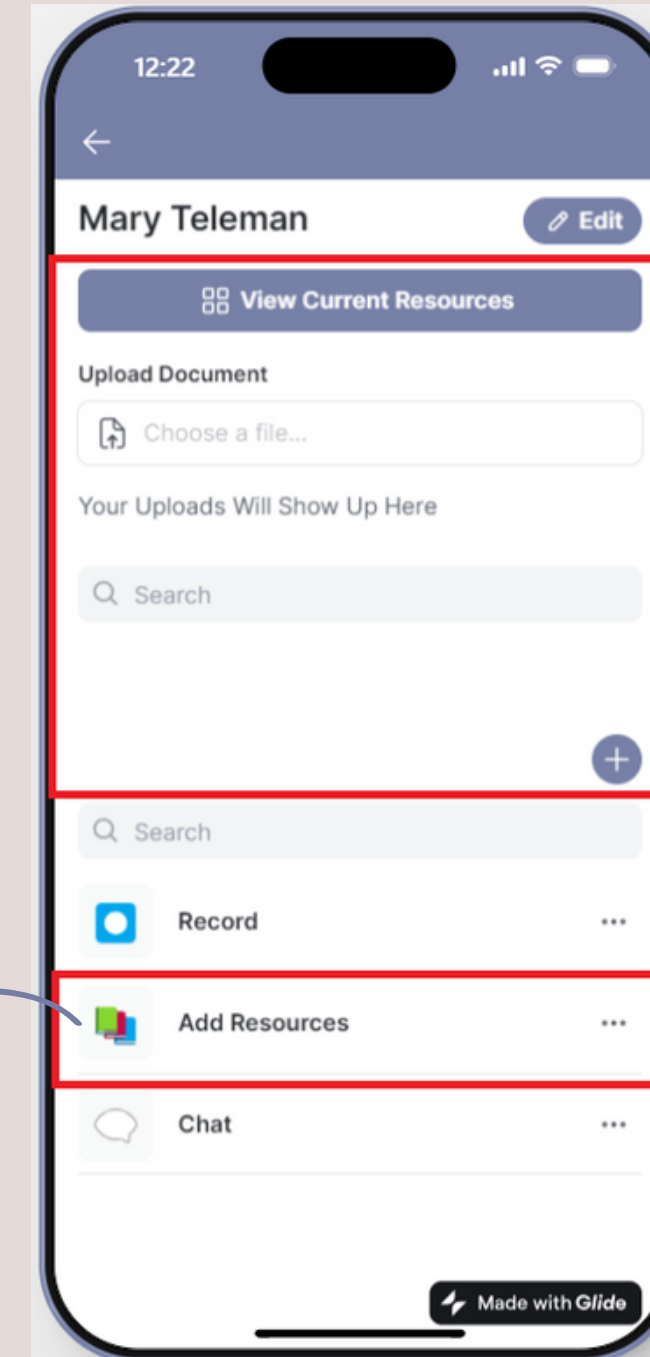
The **student's** captions side.

Resources

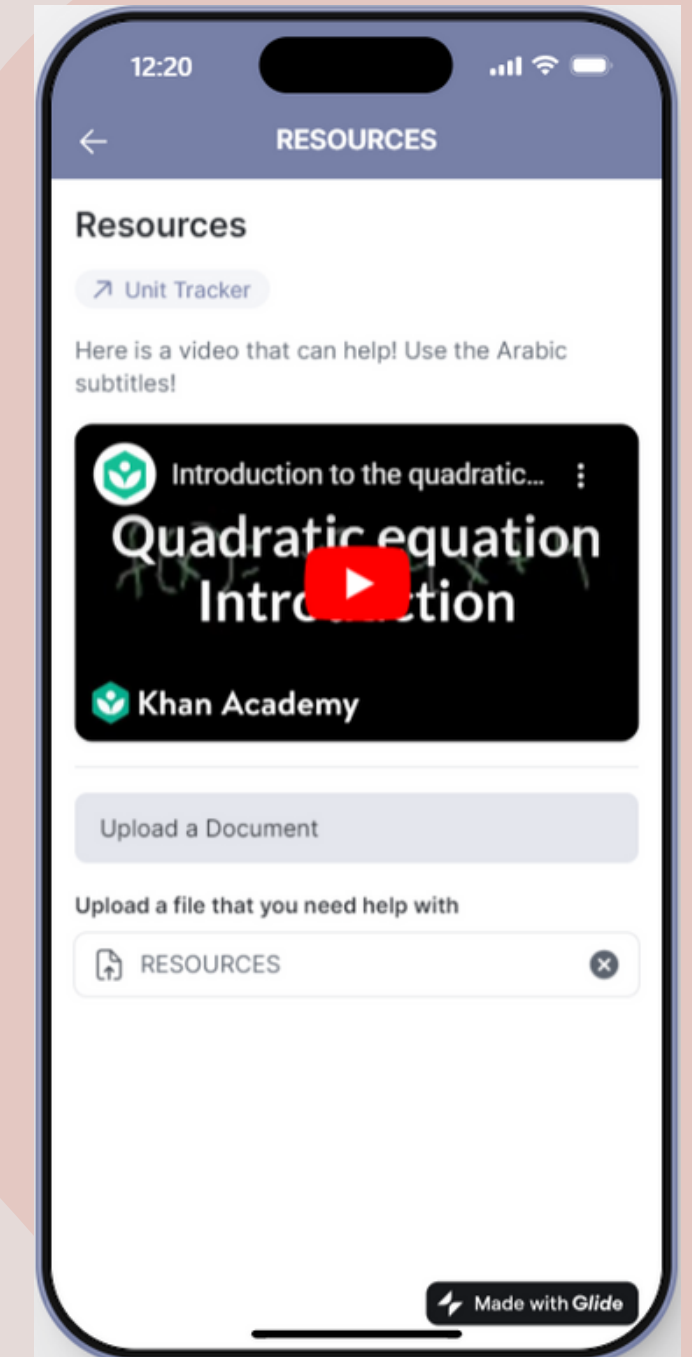
As non-native English speakers, students might need **additional/ alternative resources**. In order to make sure that teachers and the students can **upload** and **access** these materials efficiently, we decided to dedicate sections for just this. The teacher can upload explanations, videos, links, and other documents, which the student will be able to access in both English and their native language.



A mobile app form titled "Add item" with a close button (X) in the top right corner. It contains three input fields: "Label", "Upload" (with a "Choose an image..." button), and "Link". At the bottom, there are "Cancel" and "Submit" buttons.



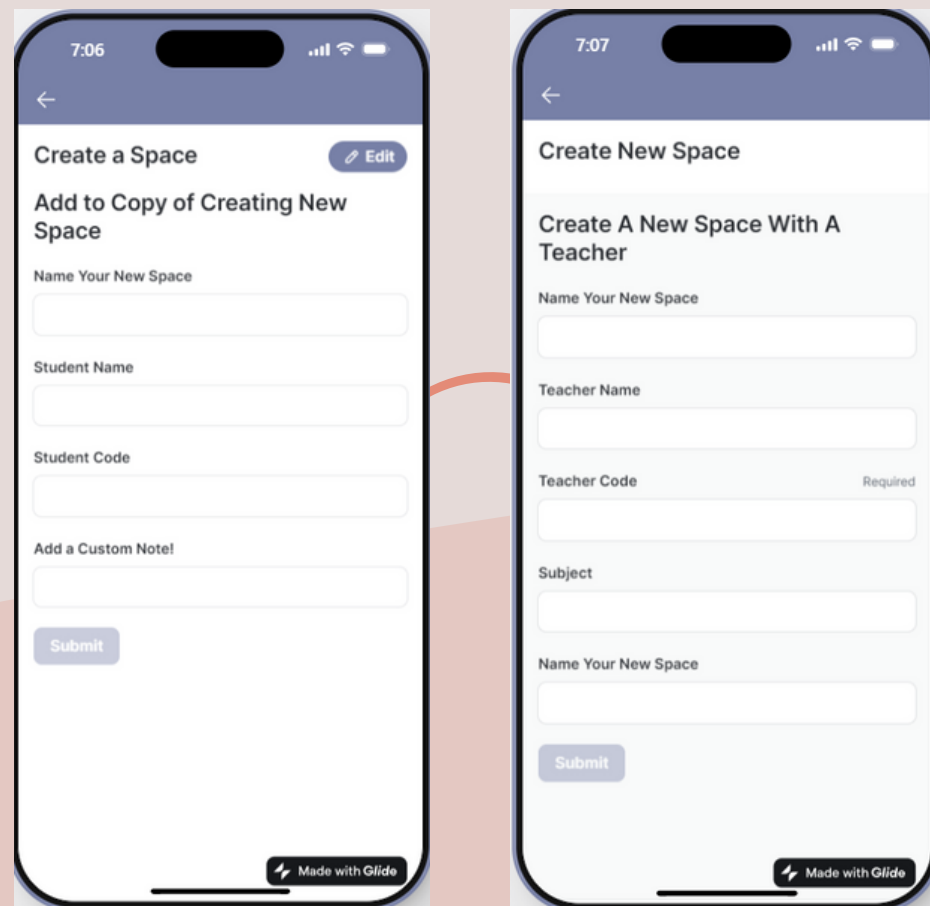
The **teacher's** resources viewpoint.



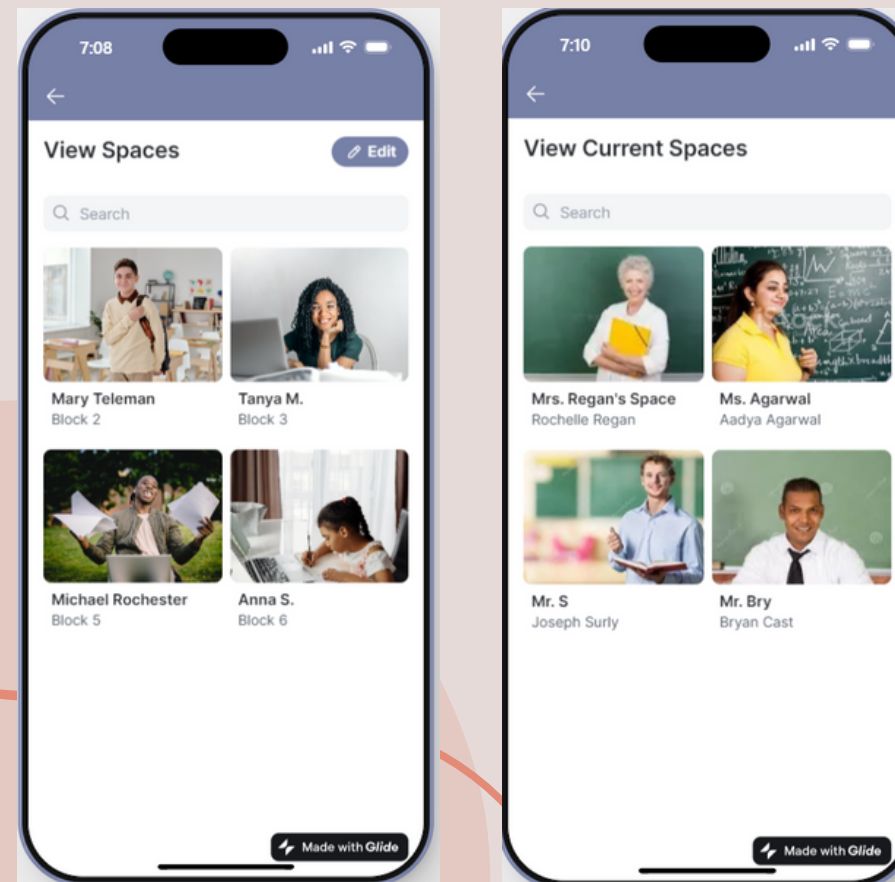
The **student's** resources viewpoint.

Additional Features

Students and teachers can create a new “space” by using user-specific codes.



Students and teachers can have a grid-view of all the teachers/students they have a space with currently.



Further Insights

- We plan to integrate more AI into our app by:
 - Voice to text transcription (live class captions)
 - Creating digital assistant to guide the users through the app
 - AI voice translations
- We also want to address how different words have different meanings and implications in different language and cultures. This is a topic we want to address in the future, and develop a feature to fix this problem.

Conclusion

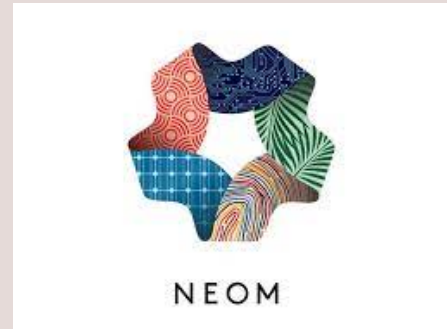
Language barriers can negatively affect students' academic performance because it means that students do not completely understand the material being taught. Many academic resources available online also cater to mainly English speakers.

Our original idea was to create an app that would help learn English or other languages itself. However, we noticed that we already have excellent apps to learn languages, such as Duolingo.

After this reflection, our team had the idea to create an app to overcome the language barrier for everyone and for every language. And then, we created the final solution: the app that could connect students and teachers from around the world and remove the language barrier, specifically in academic settings.

Our app aims to target all non-native English speaker students and create a medium for them to connect with their teachers and use the features of our app to bridge any gap of understanding that might arise from a language barrier.

Bibliography and Acknowledgments



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Thank you!